Group Project Final Report

Interruptible MLE Agents: Human Intervention in Long-Horizon Tree Search

Group ID: Group 7

DSA4213: Natural Language Processing for Data Science
Semester 1, AY2026/2027

Administrative Information

Group members	Nabil Rakaiza Abror (A0305715J) Dziand Dafi Ginandjar (A0354009J) Medhini Karthikeyan (A0351868N) Sahishnav Kritin Karamarathur Senthil Kumar (A0357345X)
Project direction	Human-in-the-loop, tool-using agents for machine learning engineering.
Repository	https://github.com/nabilrakaiza/interruptible-mle-agent

Project Summary

MLE agents steer a long search with a validation score they compute themselves, and prior work shows this score drifts away from held-out performance. We built a tree-search MLE agent that a human can interrupt: it shows its tree, its findings and a hidden *probe* score on a dashboard, and applies structured interventions at safe points between nodes. A separate evaluation agent (A1) writes the splits and scoring harness once, and the coding agent cannot change them. We ran 118 four-hour runs on six MLE-bench Lite tasks (text, tabular, image), comparing an autonomous baseline (C1: the full agent with no monitor), two scripted policies that stand in for a monitoring human, and two ablations, and analysed the 63 runs on the three text tasks in depth. No condition beat C1 (Friedman $p = 0.55$ over task $\times$ seed blocks on the text tasks; no task separates them). With A1 in place, the drift the interventions target barely occurred; without it, validation was more optimistic on the text tasks and nomad, and 2 of 17 runs leaked. The differences between tasks are consistent with the size of the held-out splits. Five live human sessions changed no outcome beyond noise; as a design implication, they suggest a person's input would matter most as task knowledge and a direct instruction channel rather than score-watching.

1. Problem, Research Question, and Motivation

> **Which human interventions into a tree-search MLE agent improve its outcomes, and at what interaction cost?**

Two sub-questions follow: *can a human, given a progress view, detect and correct the validation–test divergence that the agent cannot see?*, and *does it matter whether an intervention changes the agent's next action* (prune, redirect, override) *or its beliefs* (inject knowledge, correct a stored finding)?

Motivation. Toledo et al. [3] found that an agent's cross-validation score keeps climbing over a 24-hour run while its test score plateaus, and that picking the final node by test instead of validation would raise the medal rate by 9 to 13 points. The failure is invisible from inside the loop: in MLE-STAR [4], validation accuracy on `spaceship-titanic` rose from 0.819 to 0.868 while test accuracy fell from 0.803 to 0.734 because generated features used the target. Yet the published MLE agents [2, 3, 4, 5, 6] are autonomous by construction and give a human no way in. An agent-based approach is needed because the search space is the space of programs, with no predefined space for AutoML [2]: the system interleaves reasoning with execution feedback [9], searches candidates with a verifier [10] and acts through a constrained interface [11], and the question arises only because it makes hundreds of consequential decisions unsupervised. The intended setting is a practitioner who leaves an MLE agent running for hours and wants to oversee it, not do the work by hand.

Success meant an interruptible agent with valid submissions on most tasks, a documented intervention vocabulary, a controlled comparison and reproducible artefacts. The first three were met and the fourth in part: run folders contain competition data, so only their summaries are public. Beating the state of the art was not a goal.

Changes since the proposal. Each member ran the plan on their own task(s): the full 21-run plan on five tasks and a 13-run half plan on cactus, with insults and pizza on Kaggle CPUs without Docker. C4's thresholds came from pilot replays and C5's were fixed in advance, so the live sessions (none on pizza) were compared against the rules instead of used to write them. To spend compute and time on seeds we did not run A2-off, the exploration-diversity metric or the contamination spot-check, and A1's leak screen uses static rules instead of an LLM judgement. Two LLM-monitor runs sit outside the study (Appendix F).

2. Related Work and Contribution

MLE-bench [1] turns 75 Kaggle competitions into offline tasks with the original medal thresholds; we use its 22-task Lite subset. AIDE [2] searches over scored Python scripts grown by draft, debug and improve operators. AIRA-dojo [3] varies these components one at a time and documents the proxy-metric overfitting above. MLE-STAR [4] adds web retrieval of candidate models and leakage checkers. ML-Master [5, 6] adds MCTS and a hierarchy of distilled findings, close to the reflections of Generative Agents [12]. For general agents, AGDebugger [7] adds rollback and message editing, and Magentic-UI [8] co-planning and action approval; neither handles a search over scored programs, where interruption must respect long-running training jobs.

Contribution. The tree and operators follow AIDE, the findings ML-Master 2.0, and the leakage screen MLE-STAR. Beyond these, we contribute (1) **interruptibility as a property of the search loop**: safe-point checkpoints, an intervention queue shared by humans and scripted policies, a dashboard with a hidden probe score, and run replay; (2) **a measurement of intervention value in MLE** over 118 controlled runs on six tasks plus five live human sessions; (3) **structural separation of evaluation from optimisation**, enforced by the environment, stress-tested and ablated; and (4) **a well-supported negative result** with an explanation (Section 6). We use MLE-bench's data preparation and graders, DeepSeek V4 Flash 0731 [14] through OpenRouter [15], Docker [16], Streamlit [17], and scikit-learn [18] and PyTorch [19] in the sandbox.

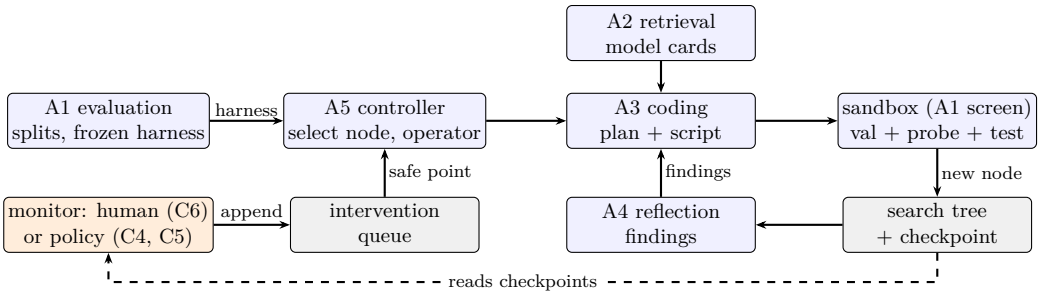


Figure 1: System overview. The agent process is the only writer to the tree. The dashboard and scripted policies run as separate processes that read checkpoints and append to the queue, so human and scripted interventions take the same code path.

3. Agent and Task Setup

Task and environment. Given a competition description and data, the system writes `submission.csv`, graded by MLE-bench. Each run has 4 hours, 2 CPUs, a 20-minute limit per script and a 200-node cap. Scripts ran CPU-only in a Docker sandbox (4 GB, no network) on each member's laptop, or in a subprocess sandbox on Kaggle for insults and pizza (machines in Appendix J). All LLM calls use DeepSeek V4 Flash 0731 (fp8) at temperature 0.7.

Sub-agents. A1 (evaluation) runs once: it splits the labelled data into training, validation and a 10% *probe* split with a fixed seed, so all runs on a task share identical splits, writes the scoring harness and freezes it. Before execution it screens each script with static rules for reading outside the input folder, the target among the features, statistics fitted on prediction rows, and validation rows used for fitting. **A2 (retrieval)** returns up to 8 model cards from allowlisted library documentation; solution sites are blocked. **A3 (coding)** implements draft, debug and improve, each a plan plus a self-contained script that trains, scores itself through the harness and predicts the test set. **A4 (reflection)** writes up to 3 findings every 10 nodes, each citing its nodes and scores; the 10 newest are shown to A3. **A5 (controller)**, not an LLM, selects the next node and operator, enforces stopping rules, checkpoints and applies interventions; it is the only writer to the tree. Permissions are enforced by the environment, not prompts (Appendix I): A3 can neither modify the harness nor reach the network. The probe score is never shown to A3 or used by A5; every node's test score is graded for analysis but never shown to any agent. **Interruption semantics.** A *safe point* is a node boundary, after a script has been scored and before A5 selects the next node. There A5 checkpoints the tree, findings, harness, budget and random state, and drains an intervention queue: *action* interventions (**prune** a subtree, **redirect** the next node and operator, **override** the final pick), *belief* interventions (**inject** a model card, **correct** or delete a finding) and **rebudget**. The clock keeps running while a human thinks, so all runs get equal time, and every action is logged with its time, target and note. **Stopping**: a run ends at the budget or node cap; a debug chain stops after 3 failures and an improve chain after 3 attempts without a 0.1% relative gain.

ID	Condition	What changes	Seeds
C1	Autonomous	Greedy search, best-validation submission.	0–4
C2	Oracle pick	C1's best node on test: upper bound on selection.	post hoc
C3	Top-3	Best of C1's top 3 nodes by validation [3].	post hoc
C4	Divergence prune	*Action.* Prune a lineage once validation has improved by $\geq$1% while the probe got $\geq$1% worse over $\geq$2 nodes since its best-probe node, in $\geq$95% of paired bootstrap resamples.	0–4
C5	Finding correction	*Belief.* Delete a finding once half of its cited nodes gained more than 0.02 more on validation than on the probe.	0–4
C6	Live human	A group member watches one live 2-hour run on the dashboard and can intervene; one session per task, none on pizza.	100
A1-off	No evaluation agent	A3 writes its own split and metric.	0–2
A4-off	No reflection	No findings in A3's prompt.	0–2

Table 1: Conditions. C4 and C5 are scripted monitors that use the same queue as a human.

4. System and Study Design

Main loop (Figure 1). A5 picks a node and operator greedily on validation (5 drafts, 50% debug chance, 15% random improves); A3 writes a script; A1 screens and scores it; A5 applies queued interventions. The best validation node is submitted.

Design choices. A1's harness is immutable to A3 by file permission, so the agent cannot score itself on data it trains on unless A1 is switched off. The probe is hidden so the monitor has a signal the agent never optimises against: a gap between validation and probe can reveal overfitting the agent cannot see. Scripted policies stand in for a human because they are reproducible across seeds, whereas a recorded human action points to a different node on any rerun of a nondeterministic agent.

Conditions (Table 1) change one component each. C2 and C3 need no runs: every node keeps its graded test score, so both come from C1's trees.

5. Evaluation and Results

Protocol. Six MLE-bench Lite tasks: spooky author identification (log loss, lower is better), insults and pizza (AUC) on text, nomad (RMSLE) on tabular data, and leaf (log loss) and cactus (AUC) on images. Each full-plan task has 21 runs (C1, C4, C5 on seeds 0–4; A1-off, A4-off on seeds 0–2) and cactus 13 (seeds 0–2 and 0–1), all at 4 hours with the same model and CPUs: 118 runs. We report MLE-bench's outcomes (valid, above median, medal) and test score, and three process metrics: the pick's validation–test gap, probe fidelity (rank correlation of probe and test over a run's nodes) and selection loss (pick versus the run's best node on test). With 3 to 5 seeds per cell we do not rank conditions whose intervals overlap; across tasks we use a Friedman test with task $\times$ seed blocks. The in-depth analysis uses the 63 text-task runs, whose run folders we hold; for the others we use each owner's summaries.

Outcomes. All 63 runs made a valid submission above the median (Table 2): insults is gold on 21 of 21 runs, pizza bronze on 9, spooky bronze on 1. Table 3 answers each claim the proposal tied to a comparison.

Condition	Spooky ↓	Insults ↑	Pizza ↑	Medals	Rate [95% CI]
C1	0.3290 ± 0.033	0.9019 ± 0.005	**0.6978** ± 0.010	9 of 15	60% [33, 87]
C2 (oracle)	0.3284 ± 0.034	0.9078 ± 0.001	0.7037 ± 0.007	10 of 15	67%
C3 (top 3)	0.3288 ± 0.033	0.9019 ± 0.005	0.6983 ± 0.009	9 of 15	60%
C4	0.3345 ± 0.027	0.9054 ± 0.004	0.6851 ± 0.019	7 of 15	47% [20, 73]
C5	**0.3133** ± 0.023	0.9074 ± 0.006	0.6847 ± 0.011	7 of 15	47% [20, 73]
A1-off	0.3592 ± 0.043	0.9062 ± 0.006	0.6851 ± 0.013	4 of 9	44% [11, 78]
A4-off	0.3424 ± 0.019	**0.9107** ± 0.002	0.6818 ± 0.026	4 of 9	44% [11, 78]
Median / bronze	0.4188 / 0.2938	0.7784 / 0.7911	0.5996 / 0.6921		

Table 2: Mean ± SD of test score over seeds per condition and task; medals pooled over tasks, with bootstrap intervals over runs. Per-task medals are in Appendix A.

Claim (proposal)	Evidence (spooky / insults / pizza)	Verdict
The agent overfits its own proxy metric	Pick's test minus validation with A1 (median): −0.006 / −0.001 / −0.021	Only pizza
The probe is a usable signal	Probe–test rank correlation 0.97 / 0.59 / 0.28 (validation: 0.98 / 0.55 / 0.53)	Depends on task
A human-style policy corrects drift	C4 within seed noise of C1 on every task; its rule never changed a pick; C2 caps any selection gain at 0.0006 / 0.006 / 0.006	No
Judgement beats a blind hedge	C3 gains at most 0.0005; C4 is no better beyond noise	No
Action and belief differ	C4 fired live on 2 of 15 runs; no C5 correction changed a pick	Not testable
Evaluation separation matters	A1-off gap −0.088 / −0.015 / −0.045; 2 of 17 A1-off runs leaked (all tasks)	Yes
Accumulated findings help	A4-off best on insults, worst on pizza; 61% of findings supported	Not shown
Interventions are worth their cost	No C4/C5 action changed a pick; 29 human actions, no gain beyond noise	Not testable

Table 3: Each claim from the proposal's evaluation plan, with the evidence and our verdict. Gaps are the pick's test minus validation score, oriented so that negative means validation was optimistic.

5.1 Comparisons and Ablations

No condition is clearly better. The best condition differs by task (C5 on spooky, A4-off on insults, C1 on pizza), and one condition's own seeds vary more than the gaps between conditions (spooky: 0.037 to 0.082 within a condition, at most 0.046 between means). Friedman tests find no difference among C1, C4 and C5 ($\chi^2 = 1.2$, $p = 0.55$, 15 task × seed blocks) or among all five run conditions ($\chi^2 = 2.2$, $p = 0.69$, 9 task × seed blocks). The strongest hint, pizza's C1 (4 bronze of 5 against 5 of 16), favours no intervention.

Selection costs little (C2, C3). The pick is a median 0.000, 0.003 and 0.009 below the run's best node on test; the top-3 hedge recovers at most 0.0005, and picking by the probe is never better on average (Appendix B).

The interventions rarely fired. Live C4 fired only on pizza (3 prunes in 2 runs); live C5 made 13 corrections (2 spooky, 11 pizza). None changed a final pick. Replayed on the 48 runs without C4, the C4 rule fires on 11 (8 pizza, 3 spooky): the pick improves on 3, is unchanged (within 0.0001) on 3 and gets worse on 5, by up to 0.024.

Task	Data	Val / probe	Valid	Medals	Best	C2 gain	Live C4 / C5
Spooky	text	3,524 / 1,762	21/21	1 bronze	C5	0.0006	0 / 2
Insults	text	789 / 395	21/21	21 gold	A4-off	0.006	0 / 0
Pizza	text, tab.	576 / 288	21/21	9 bronze	C1	0.006	3 / 11
Nomad	tabular	432 / 216	17/21	15 silver, 2 bronze	A4-off	0.00044	2 / 2
Leaf	image feat.	178 / 89	21/21	1 gold, 2 silver, 2 bronze	C1	0.077	not logged
Cactus	image	2,835 / 1,418	13/13	none (needs 1.0)	A1-off	0.0000	0 / 0

Table 4: All six tasks at 4 hours. "Best" is the condition with the best mean test score; in no task does it separate from the others. "C2 gain" is the oracle pick's mean gain over C1's validation pick. Live C4 / C5 counts prunes and finding corrections. Split sizes for nomad and cactus are approximate (20% and 10% of the training rows).

A1 keeps validation honest (A1-off). Without A1 the pick's validation is more optimistic on every text task (median gap -0.088, -0.015, -0.045 against -0.006, -0.001, -0.021), and 2 of the 17 A1-off runs across all tasks leaked (spooky and leaf; Section 6). A1-off changes more than the screen: the agent also writes its own split and metric, so the ablation tests the whole separation. With A1 no text-task run leaked, and on all 18 spooky runs the pick's test stays within 0.015 of validation. A1's screen stopped 3 real scripts that trained on validation rows, and in a stress test it flagged 4 of 4 planted leaks, one per rule, and 0 of 4 clean controls.

The other three tasks (Table 4; Appendices G–H) repeat the pattern: no condition separates. On nomad 4 runs produced no working node (likely Docker/WSL2 timeouts) and all 17 valid runs medalled. Without A1, nomad's validation was again more optimistic (pick gap -0.0047 against -0.0004) and ranked nodes almost unlike test (rank correlation 0.09 against 0.67). Cactus is a ceiling task (every node near AUC 0.9999, a medal needs 1.0); A1-off did not leak, and A1's screen stopped 4 late stacking scripts fitted on validation rows. Leaf is the one task where selection lost much: C1's mean log loss is 0.085 against 0.0075 for C2, with 178 validation rows for 99 species. Leaf's A1-off run at 15.66 leaked: the agent fitted a 99×99 calibration matrix on its own validation rows and scored it on them (validation loss 0, probe 19.44, test 15.66). A1's static screen misses this (the fit is a `scipy` optimisation, not a `.fit` call), but with A1 on the frozen harness scores validation itself; replayed, the C4 rule prunes the leaking branch at n9 and recovers test 0.055.

6. Analysis of Results, Failures, Limitations, Costs, and Risks

The differences between tasks are consistent with the size of the held-out splits. Spooky has 3,524 validation and 1,762 probe rows, insults 789 and 395, pizza 576 and 288. Where the splits are large, validation and probe rank nodes almost like test (spooky 0.97), the pick is the best node, and the rules have nothing to act on. Where they are small (pizza), both are noisy, the probe is worse than validation (0.28 against 0.53), the pick loses most, and the rules fire on noise. One case where an intervention did not help: on C5's pizza seeds 2–4, the policy deleted 5 findings that test showed were right. Pizza also shows the optimism of [3] on a small scale: the current pick's validation exceeds its test by 0.016 a quarter into the run and 0.021 at the end, since the more nodes compete on 576 rows, the more the winner's score is luck. On spooky the gap does not grow. The other tasks fit: leaf, with the smallest splits (178 and 89 rows), is where selection lost most; cactus, saturated near AUC 1.0, and nomad, a regression task whose continuous targets carry more

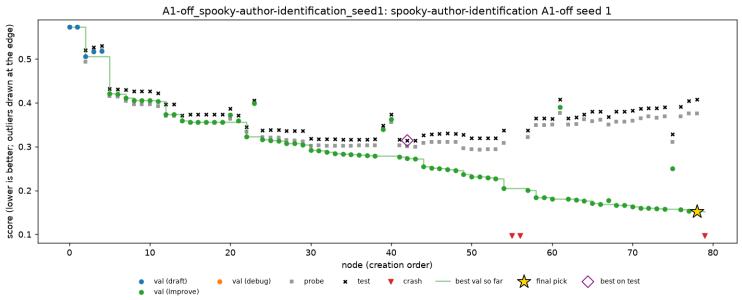


Figure 2: Spooky A1-off seed 1 (log loss): validation (green) falls; probe (grey) and test (black) rise.

information per row, gave the rules little to act on. Size is not the whole story: shrinking spooky's splits to pizza's size raises its median selection loss from 0.000 to 0.002 but leaves probe fidelity at 0.93, far above pizza's 0.28. In our data, the drift we set out to catch appeared only where held-out data was small, which is also where the monitor's signal was too noisy.

Case study: a leak. In spooky A1-off seed 1 the agent stacked models on its own validation split (Figure 2). Validation fell to 0.152 while probe and test rose after node 44, ending at 0.408. This is the divergence C4 looks for: replayed on this run, the rule would have improved the pick to 0.314. With leaf's A1-off run, it is one of the rule's two clear wins, both leaks that A1 prevents by construction.

The model family matters more than the condition. The 9 best of spooky's 21 runs on test all submitted naive Bayes combined with a linear model; on insults, refitting boosted trees on the exact features of 9 top nodes (top 3 by validation, probe and test over the 18 A1-on runs) made all 9 worse; pizza's best runs are tree ensembles. Which family a run lands on early separates runs more than any condition.

Failure modes. Of 5,381 nodes, 93.6–97.3% per condition ran; the rest are runtime errors, timeouts and 38 generation errors with no script, hence 5,343 scripts (Appendix D). The larger failure is quiet: 63% of working improve nodes did not beat their parent and 158 changed no prediction. Of 1,007 findings that claim an effect, only 61% are supported by the scores they cite (22–50% hold on the probe), yet all are shown to the agent as true, which may be why A4-off does no worse.

What a person added (C6). Five 2-hour sessions ran, one per task except pizza (Appendix E). No session beat C1's range. People read the dashboard sensibly: one spooky prune matched the C4 rule exactly, but most prunes removed weak branches (far behind with little time left, broken, or the wrong model family), which the rule does not look at. Redirects and injects did not produce the intended change: a redirect chooses which node is worked on, not what changes, and an injected card was ignored for 20 nodes. On leaf the participant was unsure when to act and made no intervention. As a design implication rather than a finding, the sessions point to where a person could add value: task knowledge (spooky stalled without the naive Bayes recipe an expert would supply), data checks (pizza's `giver_username_if_known` column leaks the label, invisible to A1) and a direct instruction channel.

Costs. About $21 for the 63 text-task runs ($0.29–0.36 per run), $4.24 for nomad's 21 and $1.63 for cactus's 13; the policies added no measurable cost, and the five C6 sessions took 29 human

actions over 10 person-hours (about $0.12 of API calls each) for no gain beyond noise. Seed 0 ran slower, before we dropped a provider that cut off 9% of its calls.

Limitations and risks. (1) Five seeds for C1, C4 and C5 and three for the ablations; [3] recommends ten, as rankings can flip at five. (2) 4 hours instead of 24; [3] reports that agents diverge only after about 15 hours, and half of spooky's runs found their pick in the last 3% of nodes. (3) One model, and the in-depth analysis covers only the text tasks; on nomad 4 runs failed on an unstable machine. (4) Scripted policies are reproducible but not behaviourally faithful, and C6 is five sessions by group members who knew the agent but were not experts in the tasks, none on pizza: illustrative, not a user study; an expert might judge more reliably. (5) Insults and pizza ran without container isolation. (6) Pizza's data leaks the label (MLE-bench recommends dropping it); no pick used that column. (7) Spooky seeds 0–2 were resumed from 2 to 4 hours. (8) None of the 5,343 text-task scripts loaded pretrained weights or downloaded anything, so every model was trained from scratch on 2 CPUs, a large part of the gap to published agents (Appendix C). The system executes generated code, A1's static screen misses leaks it has no rule for (as on leaf) and any leak in the data itself, and A4's unreliable findings can steer the search.

7. Reproducibility and Responsible Practice

Reproduction. The repository's `README.md` gives setup and launch instructions; each condition is a YAML file in `configs/conditions/` over `configs/base.yaml`. The model is pinned (`deepseek/deepseek-v4-flash-0731`, fp8, temperature 0.7, provider routing in the config), and A1's split seed is fixed, so all A1-on runs on a task share identical splits and harness. Every prompt, completion and checkpoint is logged; the repository holds each run's grade, outcomes, intervention log and analysis tables, and `docs/results/analysis/` regenerates the tables. Raw run folders contain competition data, so they stay with the members who ran them, available on request. LLM outputs are not deterministic, so our claims are distributional.

Safeguards. Scripts run in a Docker sandbox with no network, credentials or host access, except the Kaggle subprocess sandbox noted above. A2 returns only a fixed card schema from allowlisted documentation, labelled untrusted in A3's prompt [13]; a planted instruction on an allowlisted page never reached generated code (0 of 8 drafts over two models), which shows the defences held here, not that injection is impossible. API keys stay in environment variables; C6 participants were group members and no personal data was collected. MLE-bench tasks are public, which costs twice: A2 must block solution sites, and the model may have seen solutions in training; tasks whose solutions are not public would avoid both.

8. Conclusion

Over 118 controlled runs on six tasks, interventions modelled on a monitoring human did not improve an interruptible tree-search MLE agent: the validation–test drift they target barely occurred once evaluation was separated from optimisation, and where held-out data was small enough for drift to appear, the monitor's signal was too noisy to act on. Without that separation, 2 of 17 runs leaked. As a design implication, a person may help most by supplying task knowledge, checking the data, and telling the agent what to change.

Contribution Statement

Member (Student ID)	Main contributions	Share (%)
Nabil Rakaiza Abror (A0305715J)	Proposed the project idea and research question; wrote the proposal and work plan and coordinated the role split and run schedule. Built the core agent (A3 coding, A5 controller: operators, greedy search, safe points, checkpoints and resume, final selection) and A2 retrieval with the prompt-injection red team. Integrated all roles (sandbox limits, MLE-bench grading, grid launcher, cost and outcome reports, policy replay, merged dashboard, C6 protocol). Ran the spooky, insults and pizza full plans (63 runs) and the spooky and insults C6 sessions; did the cross-task analysis and compiled all results.	50
Dziand Dafi Ginandjar (A0354009J)	Built A1 (seeded splits, frozen harness, leakage screen, A1-off mode), A4 reflection with evidence-backed findings, and the C5 finding-correction policy; wrote the leakage stress test, finding-accuracy check, and validation–test gap and probe-fidelity analyses. Ran cactus at 2 and 4 hours (26 runs) and its C6 session, two runs with an LLM monitor (cactus, pizza), and fixed pizza's CSV quoting.	20
Medhini Karthikeyan (A0351868N)	Built the Docker execution sandbox, LLM call logging and pilot-run infrastructure, and the ML Docker image. Ran leaf's half plan (13 runs) and full plan (21 runs at 4 hours), its C6 session, grading and write-ups.	15
Sahishnav Kritin Karamarathur Senthil Kumar (A0357345X)	Built the C4 divergence-prune policy and the first version of the monitoring dashboard; fixed the coding agent's `OUTPUT_DIR` filtering. Ran nomad's half plan (13 runs) and full plan (21 runs at 4 hours), its C6 session, and the write-ups.	15
Total		**100**

References

[1] J. S. Chan et al. MLE-bench: Evaluating Machine Learning Agents on Machine Learning Engineering. *ICLR*, 2025. arXiv:2410.07095.

[2] Z. Jiang, D. Schmidt, D. Srikanth, D. Xu, I. Kaplan, D. Jacenko, and Y. Wu. AIDE: AI-Driven Exploration in the Space of Code. arXiv:2502.13138, 2025.

[3] E. Toledo, K. Hambardzumyan, M. Josifoski et al. AI Research Agents for Machine Learning: Search, Exploration, and Generalization in MLE-bench. arXiv:2507.02554, 2025.

[4] J. Nam, J. Yoon, J. Chen, J. Shin, S. Ö. Arık, and T. Pfister. MLE-STAR: Machine Learning Engineering Agent via Search and Targeted Refinement. arXiv:2506.15692, 2025.

[5] Z. Liu, Y. Cai, X. Zhu et al. ML-Master: Towards AI-for-AI via Integration of Exploration and Reasoning. arXiv:2506.16499, 2025.

[6] X. Zhu, Y. Cai, Z. Liu et al. Toward Ultra-Long-Horizon Agentic Science: Cognitive Accumulation for Machine Learning Engineering. arXiv:2601.10402, 2026.

[7] W. Epperson et al. Interactive Debugging and Steering of Multi-Agent AI Systems. *CHI*, 2025.

[8] H. Mozannar et al. Magentic-UI: Towards Human-in-the-loop Agentic Systems. Microsoft Research, 2025.

[9] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao. ReAct: Synergizing Reasoning and Acting in Language Models. *ICLR*, 2023.

[10] S. Yao, D. Yu, J. Zhao, I. Shafran, T. L. Griffiths, Y. Cao, and K. Narasimhan. Tree of Thoughts: Deliberate Problem Solving with Large Language Models. *NeurIPS*, 2023.

[11] J. Yang, C. E. Jimenez, A. Wettig, K. Lieret, S. Yao, K. Narasimhan, and O. Press. SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering. *NeurIPS*, 2024.

[12] J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein. Generative Agents: Interactive Simulacra of Human Behavior. *UIST*, 2023.

[13] E. Wallace, K. Xiao, R. Leike, L. Weng, J. Heidecke, and A. Beutel. The Instruction Hierarchy: Training LLMs to Prioritize Privileged Instructions. arXiv:2404.13208, 2024.

[14] DeepSeek-AI. DeepSeek V4 Flash (`deepseek-v4-flash-0731`). Large language model, accessed through OpenRouter, 2026.

[15] OpenRouter. Unified API for large language models. https://openrouter.ai, accessed November 2026.

[16] D. Merkel. Docker: Lightweight Linux Containers for Consistent Development and Deployment. *Linux Journal*, 2014(239), 2014.

[17] Streamlit. An open-source Python framework for data apps. https://streamlit.io, accessed November 2026.

[18] F. Pedregosa et al. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

[19] A. Paszke et al. PyTorch: An Imperative Style, High-Performance Deep Learning Library. *NeurIPS*, 2019.

[20] Anthropic. Claude Opus 5.5. Large language model, 2026.

Appendix

A. Medals per task

Condition	Spooky	Insults	Pizza
C1	0 of 5	5 gold of 5	4 bronze of 5
C4	0 of 5	5 gold of 5	2 bronze of 5
C5	1 bronze of 5	5 gold of 5	1 bronze of 5
A1-off	0 of 3	3 gold of 3	1 bronze of 3
A4-off	0 of 3	3 gold of 3	1 bronze of 3

B. Other rules for the final pick

Rule	Spooky test (loss)	Insults test (loss)	Pizza test (loss)
Best validation (used)	0.3284 (0.0009)	0.9059 (0.0039)	0.6879 (0.0105)
Best probe	0.3296 (0.0021)	0.9057 (0.0041)	0.6790 (0.0195)
Validation and probe, row-weighted	0.3285 (0.0011)	0.9060 (0.0038)	0.6868 (0.0117)
Mean rank on validation and probe	0.3286 (0.0012)	0.9059 (0.0039)	0.6840 (0.0145)

Table 5: Means over the 18 A1-on runs per task. "Loss" is the distance to the run's best node on test.

C. Comparison with published agents

From the official MLE-bench grading reports of the leaderboard experiments, mean test score per task. Budgets, hardware and models differ, so this only places our numbers.

Agent	Spooky ↓	Insults ↑	Pizza ↑
Ours (21 runs, 4 h, DeepSeek V4 Flash)	0.333	0.906	0.688
AIDE, GPT-4o, CPU only	0.498	0 of 12 valid	0.650
AIDE, o1-preview	0.321	0.905	0.656
AIRA-dojo, o3	0.247	0.949	0.629
Top entries (Gemini 3 Pro)	∼0.21–0.25	∼0.955	∼0.70

D. Node outcomes by condition

Condition	Working	Runtime error	Timeout	Generation error	Leak stop	Pruned
C1	95.5%	2.0%	1.7%	0.7%	2	0
C4	93.6%	2.4%	1.5%	0.8%	0	23
C5	97.3%	0.7%	1.3%	0.7%	0	0
A1-off	95.4%	2.6%	1.5%	0.5%	0	0
A4-off	96.0%	1.7%	1.4%	0.7%	1	0

Table 6: 5,381 nodes over the 63 runs.

E. Live human sessions (C6)

Task	Monitor	Actions	Test (C1's range)	Effect
Spooky	person, 2 h	9 prune, 2 redirect, correct	0.368 (0.295–0.390)	none
Insults	person, 2 h	4 prune, 2 redirect, inject, override	0.9087 (0.901–0.909)	+0.001
Nomad	person, 2 h	1 prune	0.0614 (0.0599–0.0653)	none
Cactus	person + LLM, 2 h	8 prune	0.99953 (0.99917–0.99994)	none
Leaf	person, 2 h	none	0.0261 (0.0128–0.0601)	–

Table 7: Live human sessions (C6), one seed each, 2 hours. C1's range is over its seeds, read from checkpoints at 2 hours where the grid ran 4. "Person + LLM": the participant chose among prunes an LLM assistant suggested. Effect is on the final pick.

F. Runs with an LLM monitor (outside the study)

Two extra 4-hour runs, both on one member's Mac in Docker, used Claude (Opus 5.5) [20] as the monitor instead of a person: it read validation and probe scores, plans and scripts every 15 minutes and queued prunes through the same queue as the dashboard, following criteria taken from the human cactus session. These are not C6 sessions as the proposal defines them, and we do not use them in the main analysis.

Task	Actions	Test	C1's range (4 h)	Effect on the pick
Cactus (seed 102)	9 prune	0.99995	0.99953–0.99987	none
Pizza (seed 100)	79 prune	0.6671	0.684–0.709	worse

On cactus every node sits near AUC 0.9999, and the prunes removed only waste. On pizza the monitor pruned every child that scored below its parent on validation, which on 576 rows is noise: it removed the four best nodes on test and pushed the search into one line that improved on validation while falling on test. The C4 rule, replayed on the same run, would have pruned that line and scored 0.6795 (+0.012). A monitor without a sense of the task's noise can make a run worse. Write-ups: `docs/results/{cactus,pizza}/c6-4h.md`. The human sessions' notes, intervention logs and timelines are in `docs/results/<task>/`.

G. Nomad (full plan, 4 hours)

Condition	Valid	Mean test RMSLE ↓	Medals	Pick's gap (median)
C1	5 of 5	0.06089	4 silver, 1 bronze	−0.0001
C2 (oracle)	5 of 5	0.06044	4 silver, 1 bronze	–
C3 (top 3)	5 of 5	0.06078	4 silver, 1 bronze	–
C4	5 of 5	0.06165	4 silver, 1 bronze	−0.0006
C5	3 of 5	0.06146	3 silver	−0.0004
A1-off	2 of 3	0.06102	2 silver	−0.0047
A4-off	2 of 3	0.05907	2 silver	−0.0015

Table 8: Nomad, 21 runs; thresholds: median 0.0699, bronze 0.0658, silver 0.0623, gold 0.0559. Means are over valid runs only. The 4 invalid runs produced no working node. The gap is oriented as in Table 3: negative means validation was optimistic. From `docs/results/nomad/full-4h/`.

H. Leaf and cactus (4 hours)

	Leaf (log loss ↓)		**Cactus** (AUC ↑)	
Condition	**Mean**	**Medals**	**Mean**	**Medals**
C1	0.0849	1 silver, 1 bronze	0.99970	none
C2 (oracle)	0.0075	–	0.99970	–
C3 (top 3)	0.0149	–	0.99970	–
C4	0.1034	1 silver	0.99985	none
C5	0.1660	none	0.99968	none
A1-off	5.4541	1 bronze	0.99998	none
A4-off	0.2428	1 gold	0.99978	none

Table 9: Leaf: 21 runs, all valid; thresholds median 0.1083, bronze 0.0153, silver 0.0079, gold 0.0 (A4-off's 0.0 run reached it, as the original leaderboard did). A1-off's mean is driven by one run at 15.66, and three runs in three conditions scored exactly 0.34888 (and two exactly 0.36408), most likely because every A1-on run trains on the same split, so runs that end on the same deterministic model submit identical predictions; we do not have their scripts to confirm it. Cactus: 13 runs (seeds 0–2, ablations 0–1), all valid and above the median 0.9991; every medal needs AUC 1.0. From `docs/results/leaf/full-4h/` and `docs/results/cactus/4h.md`.

I. Permissions

	Read data	Write harness	Execute code	Web access	Write tree / KB
A1 Evaluation	yes	yes (once)	yes, sandbox	no	no
A2 Retrieval	no	no	no	allowlisted	model cards only
A3 Coding	yes	**no**	yes, sandbox	no	no
A4 Reflection	no	no	no	no	findings only
A5 Controller	no	no	no	no	yes
Human monitor	view	no	no	no	via queue only

Table 10: Permission matrix. A3 cannot modify the harness or reach the network, so it can neither redefine its objective nor fetch code. On Kaggle (insults, pizza) the subprocess sandbox does not enforce the network rule.

J. Machines

Spooky: MacBook Pro (M1), Docker. Cactus: MacBook Pro (M2), Docker in Colima. Nomad: Asus Zenbook A, Windows ARM64 (Snapdragon X X1-26-100), Docker on WSL2. Leaf: HP ENVY x360 13, Windows x64 (Intel Core i5-1230U), Docker Desktop. Insults and pizza: Kaggle CPU sessions, subprocess sandbox. Every run: 2 CPUs, CPU-only training.